Notes: Niederle & Vesterlund (QJE, 2007)

Do Women Shy Away from Competition? Do Men Compete Too Much?

Contents

1 Big picture

- **Question.** Do men and women with similar measured ability differ in their willingness to choose competitive compensation?
- **Core empirical object.** The outcome is not performance in a tournament. It is *selection into* a tournament when both piece-rate and tournament pay have been experienced.
- **Experimental advantage.** The design removes several confounds that exist in labor-market data: subjects face the same task, same incentives, same group composition, and the researcher observes both performance and beliefs.
- **Sample and task.** The sample contains forty women and forty men. Groups consist of two women and two men. Subjects add sets of five two-digit numbers for five minutes.
- **Core result.** Men and women perform similarly, but 73% of men and only 35% of women select the tournament for Task 3.
- **Performance is not the explanation.** The performance distributions under the piece rate and compulsory tournament almost overlap by gender. Conditional on performance quartile, men enter more often in every quartile.
- **Payoff implication.** High-performing women often choose the piece rate even when tournament entry has higher expected monetary payoff. Low-performing men often choose the tournament even when the piece rate has higher expected monetary payoff.
- **Belief mechanism.** Men are more confident about their relative performance. Thirty of forty men, but only seventeen of forty women, guess that they ranked best in their Task-2 tournament.
- **Beliefs explain part, not all.** Controlling for guessed tournament rank reduces the female effect in Task 3 from about -0.379 to -0.278, leaving most of the gap unexplained by beliefs alone.
- **Task-4 mechanism.** Task 4 asks subjects to submit their past piece-rate performance to a tournament. Because there is no new performance under competition, Task 4 controls for confidence, risk, and feedback concerns without requiring the subject to perform competitively.
- **Residual competition component.** Adding both guessed tournament rank and the Task-4 choice reduces the female effect to -0.162. The paper interprets the remaining gap as evidence of gender differences in preferences for performing in a competitive environment.
- **Economic takeaway.** Competitive institutions can distort the pool of entrants. If entry into competitive promotions, bonuses, or applications differs by gender, outcomes can differ even when underlying ability is similar.

2 Experimental design and measurement

2.1 Task environment

- Subjects solve simple arithmetic problems: adding five two-digit numbers.
- Each round lasts five minutes.
- The task is chosen because the authors expected, and then confirm, that women and men perform similarly.
- Subjects are assigned to groups of four with two women and two men.
- The mixed-gender group is important because the tournament is directly comparable across gender: men and women compete against the same type of group.

Interpretation: The design focuses on willingness to compete, not on a domain where men or women have a large ex ante skill advantage.

2.2 The four tasks

- **Task 1: Piece rate.** Subjects receive 50 cents per correct answer.
- **Task 2: Compulsory tournament.** Subjects receive $2 per correct answer if they solve more problems than the other three members of the group; otherwise they receive zero for the task.
- **Task 3: Choice for new performance.** Subjects choose whether their next performance will be paid by piece rate or tournament. If they choose the tournament, their Task-3 performance is compared with the other group members' Task-2 tournament performances.
- **Task 4: Choice for past performance.** Subjects choose whether to submit their earlier Task-1 piece-rate performance to a tournament. This decision is paid based on a past performance and therefore does not require a new competitive performance.
- **Beliefs.** At the end, subjects guess their rank in Task 1 and Task 2. They are paid $1 for each correct rank guess, with ties treated generously.

Interpretation: Task 3 is the main tournament-entry choice. Task 4 is the key control task because it preserves the monetary gamble and feedback feature but removes the act of performing under competition.

2.3 Why Task 4 is central

- Task 3 mixes several factors: expected payoff, overconfidence, risk attitudes, feedback aversion, and taste for performing under competition.
- Task 4 removes one factor: the subject does not have to perform again in a competitive setting.
- If Task 3 and Task 4 gender gaps were the same after belief controls, the gap would likely be about general risk, feedback, or confidence.
- The fact that the Task-4 gender gap largely disappears after belief controls, while the Task-3 gap remains, supports a competitive-performance interpretation.

Interpretation: The design is a decomposition exercise: Task 4 is not a perfect risk measure, but it is a powerful within-experiment comparison for the general features of a tournament choice.

2.4 Key measurement variables

- **TournamentChoice** equals one if the subject chooses the tournament in Task 3.
- **SubmitPast** equals one if the subject submits the Task-1 piece-rate performance to the tournament in Task 4.
- **Piece rate performance** is the number correct in Task 1.
- **Tournament performance** is the number correct in Task 2.
- **Tournament - piece rate** measures the change in performance between Task 2 and Task 1.
- **Guessed tournament rank** is the subject's belief about relative Task-2 rank from one, best, to four, worst.

Interpretation: The paper observes both realized ability and subjective beliefs. This is why it can separate performance, confidence, and residual willingness to compete.

3 Models and empirical specifications

3.1 Expected-payoff benchmark

Under the piece rate, a subject earns 50 cents per correct answer. Under the tournament, a subject earns $2 per correct answer if she wins. For a risk-neutral expected-payoff maximizer,

$$2 \times Pr(\text{win}) = 0.50 \quad \Rightarrow \quad Pr(\text{win}) = 25\%.$$

Interpretation: A subject should choose the tournament if the perceived probability of winning exceeds 25%, ignoring risk, feedback, and nonpecuniary costs or benefits of competition.

3.2 Observed entry decision

A useful reduced-form representation is

$$\text{Choose tournament if } E[\pi_T - \pi_P \mid \text{performance, beliefs}] + U_i^{comp} - C_i^{risk} - C_i^{feedback} \geq 0.$$

- $E[\pi_T - \pi_P]$ is the expected monetary gain from the tournament.
- U_i^{comp} is the utility or disutility from performing in a competitive environment.
- C_i^{risk} captures aversion to the tournament's payoff risk.
- $C_i^{feedback}$ captures aversion to receiving relative performance feedback.

Interpretation: Task 3 includes all terms. Task 4 removes the need to perform in a competitive environment, so it helps identify the general components from the competition-performance component.

3.3 Baseline probit for Task 3

The main regression can be written as

$$Pr(TournamentChoice_i = 1) = \Phi(\alpha + \beta Female_i + \gamma Tournament_i + \delta(Tournament_i - PieceRate_i)).$$

Interpretation: The coefficient on $Female_i$ measures the gender gap in tournament entry after controlling for realized performance and the improvement from piece rate to tournament.

3.4 Belief-augmented probit

The belief-augmented specification adds guessed rank:

$$Pr(TournamentChoice_i = 1) = \Phi(\alpha + \beta Female_i + \gamma Tournament_i + \delta(Tournament_i - PieceRate_i) + \eta GuessRank_i).$$

Because lower guessed rank means greater confidence, $\eta < 0$ means more confidence predicts entry.

Interpretation: Comparing β with and without $GuessRank_i$ measures how much of the gender gap is explained by confidence.

3.5 Full decomposition specification

The final Task-3 specification adds Task 4:

$$Pr(TournamentChoice_i = 1) = \Phi(\alpha + \beta Female_i + \gamma Tournament_i + \delta(Tournament_i - PieceRate_i) + \eta GuessRank_i + \kappa SubmitP$$

Interpretation: If $SubmitPast_i$ controls for confidence, risk, and feedback attitudes, the remaining female coefficient is interpreted as the gender gap in willingness to perform in competition.

3.6 Decomposition of the female effect

The paper's main decomposition is:

$$\text{Baseline female effect} = -0.379,$$

$$\text{After beliefs} = -0.278,$$

$$\text{After beliefs and Task 4} = -0.162.$$

Thus,

$$\frac{0.379 - 0.162}{0.379} \approx 57\%, \qquad \frac{0.162}{0.379} \approx 43\%.$$

Interpretation: About 57% of the original gender gap is explained by general factors captured by beliefs and Task-4 submission. The remaining 43% is the residual competitive-performance component.

4 Identification and empirical design

4.1 What the experiment identifies

- The experiment identifies a gender difference in choosing a competitive compensation scheme among subjects with similar observed ability.
- It identifies this gap in a controlled environment where all subjects experience both compensation schemes before choosing.
- It identifies part of the mechanism by comparing Task 3 and Task 4, and by controlling for elicited beliefs.

Interpretation: The study is not inferring competitiveness from labor-market outcomes. It directly observes the choice to enter a tournament.

4.2 Why the performance evidence matters

- If men performed much better, gender differences in entry could be rational expected-payoff responses.
- Figure I and Table I show that performance differences are small and statistically insignificant.
- The probability of winning the tournament is about 26% for men and 24% for women, not a meaningful gender difference.

Interpretation: The absence of an ability gap makes the large entry gap economically interesting.

4.3 Why performance quartiles matter

- Figure II shows men enter more than women in every performance quartile.
- Even high-performing women enter less often than low-performing men.
- This pattern rules out the simple claim that the tournament attracts the best performers regardless of gender.

Interpretation: The tournament selection rule changes who enters, not just who wins.

4.4 Why beliefs matter

- The expected value of tournament entry depends on perceived relative rank.
- Men are much more likely to believe they ranked best.
- Belief controls reduce the gender gap, confirming that overconfidence is an important channel.
- Beliefs do not eliminate the gap, so confidence is not the complete mechanism.

Interpretation: Beliefs are both an explanatory variable and a potential outcome of gendered experience with competition.

4.5 What the design cannot fully prove

- Task 4 controls for several general factors, but it may not fully capture risk aversion or feedback aversion.
- The laboratory task is simple and short; workplace competitions are repeated, multidimensional, and socially embedded.
- The sample is relatively small: forty women and forty men.
- The findings do not imply that all women avoid competition or that all men over-enter.
- The experiment cannot by itself distinguish innate preferences from preferences shaped by social norms, stereotype threat, previous experience, or expectations about mixed-gender competition.

Interpretation: The strongest conclusion is that gender differences in tournament entry exist in this controlled setting and are not explained by measured performance.

5 Tables and figures

5.1 Figure I: CDF of Number of Correctly Solved Problems; Table I: Performance Characteristics by Choice of Compensation Scheme

Read:

- **Figure I, Panel A** plots the CDF of Task-1 piece-rate performance. The male and female curves are close throughout the distribution.
- **Figure I, Panel B** plots the CDF of Task-2 compulsory tournament performance. The distributions again overlap closely.
- Average Task-1 performance is 10.15 for women and 10.68 for men; the difference is not significant ($p = .459$).
- Average Task-2 performance is 11.8 for women and 12.1 for men; the difference is not significant ($p = .643$).
- Table I shows that women choosing the piece rate solved 10.35 problems in Task 1 and 11.77 in Task 2; women choosing the tournament solved 9.79 and 11.93.
- Men choosing the piece rate solved 9.91 in Task 1 and 11.09 in Task 2; men choosing the tournament solved 10.97 and 12.52.
- The performance gain from Task 1 to Task 2 is positive for all groups, but the gain does not generate a gender explanation for entry.

Interpretation: The figure and table establish the first necessary fact: men and women have similar ability in this task. The large gender gap in tournament entry therefore cannot be dismissed as a performance gap.

Economic message: Competitive selection can differ even when objective performance is similar. This is why tournament-entry choice is economically important.

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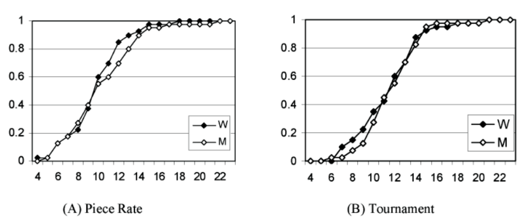

FIGURE I
CDF of Number of Correctly Solved Problems
Panel A: Piece rate (Task 1); Panel B: Tournament (Task 2)

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TABLE I
PERFORMANCE CHARACTERISTICS BY CHOICE OF COMPENSATION SCHEME (TASK 3)

		Average performance		
	Compensation scheme	Piece rate	Tournament	Tournament–piece rate
Women	Piece rate	10.35	11.77	1.42
		(0.61)	(0.67)	(0.47)
	Tournament	9.79	11.93	2.14
		(0.58)	(0.63)	(0.54)
Men	Piece rate	9.91	11.09	1.18
		(0.84)	(0.85)	(0.60)
	Tournament	10.97	12.52	1.55
		(0.69)	(0.48)	(0.49)

Averages with standard errors in parentheses. Sample is forty women and forty men.

mance between those who do and do not enter the tournament ($p \geq .35$ for each of the three performance measures). For men,

(a) Figure I: CDF of number of correctly solved problems **(b)** Table I: Performance characteristics by Task-3 compensation choice

5.2 Table II: Probit of Tournament Choice in Task 3; Figure II: Tournament Entry by Performance Quartile

Read:

- Table II reports a female marginal effect of -0.380 with $p = .00$.

- The tournament performance coefficient is 0.015 and not significant; the tournament-minus-piece-rate coefficient is also not significant.
- The marginal effect is evaluated at a man with thirteen correct answers in the tournament and twelve in the piece rate, close to the point of payoff indifference.
- Figure II shows that men select the tournament more often than women in each performance quartile.
- In both panels, women in the highest performance quartile enter less often than men in the lowest performance quartile.
- The figure is important because it uses both Task-2 performance and Task-3 performance; the conclusion is not driven by using past performance only.

Interpretation: Table II says that controlling linearly for performance does not close the gender gap. Figure II says the same thing nonparametrically by performance quartile.

Economic message: Tournaments do not simply select the highest performers. They also select people who are willing to compete, and this willingness differs by gender in the experiment.

formance under the two compensation schemes does not significantly affect the decision to enter the tournament, the participant's gender does. The reported marginal gender effect of −.380 in Table II shows that a man with a performance of thirteen in

TABLE II
PROBIT OF TOURNAMENT CHOICE IN TASK 3

	Coefficient	*p*-value
Female	−.380	.00
Tournament	.015	.41
Tournament–piece rate	.015	.50

Dependent variable: Task-3 choice of compensation scheme (1-tournament and 0-piece rate). Tournament refers to Task-2 performance, tournament–piece rate to the change in performance between Task 2 and Task 1. The table presents marginal effects of the coefficient evaluated at a man with thirteen correct answers in the tournament and twelve in the piece rate. Sample is forty women and forty men.

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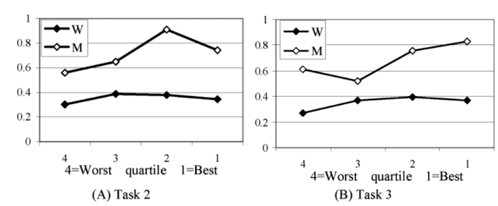

FIGURE II
Proportion of Participants Selecting Tournament for Task 3 Conditional on Task-2 Tournament Performance Quartile (Panel A) and Task-3 Choice Performance Quartile (Panel B)

(a) Table II: Probit of tournament choice in Task 3 **(b)** Figure II: Tournament entry by performance quartile

5.3 Table III: Expected Costs of Over- and Under-Entry in Task-3 Tournament

Read:

- The table classifies subjects with fourteen or more correct answers as people who should enter, and subjects with twelve or fewer as people who should choose the piece rate. Thirteen-correct subjects are treated as approximately indifferent and excluded.
- Based on Task-2 performance, twelve women and twelve men should enter. Eight of those women and three of those men do not enter.
- Based on Task-2 performance, twenty-four women and twenty-two men should not enter. Nine of those women and fourteen of those men enter anyway.
- Based on Task-2 performance, expected total cost of under-entry is 99.4 for women and 34.5 for men.
- Expected total cost of over-entry is 32.9 for women and 56.5 for men.
- Based on Task-3 performance, the same qualitative pattern appears: women have larger under-entry costs; men have larger over-entry costs.
- Total expected costs are higher for women using Task-2 performance, 132.3 versus 91.0, and remain higher using Task-3 performance, 113.5 versus 93.3.

Interpretation: Women and men make different kinds of payoff mistakes. Women more often fail to enter profitable tournaments. Men more often enter unprofitable tournaments. The paper's title reflects both sides: women shy away from competition, and men compete too much.

Economic message: Selection into tournaments can reduce allocative efficiency in both directions: high-ability potential entrants may stay out, and low-ability entrants may enter.

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TABLE III
EXPECTED COSTS OF OVER- AND UNDER-ENTRY IN TASK-3 TOURNAMENT

	Calculation based on Task-2 performance		Calculation based on Task-3 performance	
	Women	Men	Women	Men
Under-entry				
Number who should enter	12	12	9	20
Of those how many do not enter	8	3	6	4
Expected total cost of under-entry	99.4	34.5	84.6	49.6
Average expected cost of under-entry	12.4	11.5	14.1	16.5
Over-entry				
Number who should not enter	24	22	24	19
Of those how many do enter	9	14	8	12
Expected total cost of over-entry	32.9	56.5	28.9	43.8
Average expected cost of over-entry	3.7	4.0	3.6	3.6
Total expected costs	132.3	91.0	113.5	93.3

Participants solving fourteen or more problems should enter the tournament, and those with twelve and fewer problems should select the piece rate. Participants with thirteen problems (who are virtually indifferent between the two compensation schemes) are not included in the analysis.

V. EXPLANATIONS FOR THE GENDER GAP IN TOURNAMENT ENTRY

Table III: Expected costs of over- and under-entry in Task-3 tournament

5.4 Table IV: Distribution of Guessed Tournament Rank; Table V: Ordered Probit of Guessed Tournament Rank

Read:

- Table IV shows that thirty of forty men guess they ranked first, compared with seventeen of forty women.
- Only five men guess rank 2, four guess rank 3, and one guesses rank 4. Women are more spread out: seventeen guess rank 1, fifteen rank 2, six rank 3, and two rank 4.
- Both genders are overconfident relative to actual ranks, but men are substantially more overconfident.
- Many first-place guesses are wrong: twenty-two of the thirty male rank-1 guesses are incorrect, and nine of the seventeen female rank-1 guesses are incorrect.
- Table V confirms the pattern in an ordered probit. The female coefficient is 0.75 with $p = .01$; because higher rank numbers are worse, women guess worse ranks conditional on performance.
- The coefficient on tournament performance is -0.19 with $p = .00$, meaning higher performers guess better ranks.
- The tournament-minus-piece-rate coefficient is not significant, suggesting beliefs are driven more by absolute tournament performance than by improvement over piece rate.

Interpretation: Beliefs are systematically gendered. Men are not merely entering more because they perform better; they also believe they are relatively better.

Economic message: Confidence is a selection mechanism. In competitive environments, subjective beliefs can affect entry as much as objective performance.

GENDER DIFFERENCES IN COMPETITIVENESS

TABLE IV
DISTRIBUTION OF GUESSED TOURNAMENT RANK

	Men		Women	
	Guessed rank	Incorrect guess	Guessed rank	Incorrect
1: Best	30	22	17	9
2	5	3	15	10
3	4	2	6	5
4: Worst	1	1	2	1
Total	40	28	40	25

guessing that they ranked second or third.[16] However, pa
pants believe that they are ranked substantially better than
Table IV shows the participants' believed rank distribution:

(a) Table IV: Distribution of guessed tournament rank

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TABLE V
ORDERED PROBIT OF GUESSED TOURNAMENT RANK

	Coefficient	Standard error	*p*-value
Female	.75	0.30	.01
Tournament	−.19	0.06	.00
Tournament–piece rate	−.08	0.07	.27

Ordered probit of guessed rank for guesses of ranks 1, 2, and 3. Sample is thirty-eight women and thirty-nine men.

quently than women? Figure III shows for each guessed rank the proportion of women and men who enter the tournament.[18] While tournament entry-decisions are positively correlated with the

(b) Table V: Ordered probit of guessed tournament rank

5.5 Figure III: Tournament Entry Conditional on Guessed Rank; Table VI: Probit of Tournament-Entry Decision with Beliefs

Read:

- Figure III shows a strong positive relation between confidence and entry: subjects who guess better ranks enter more often.
- Yet gender differences remain within guessed-rank categories. Even among subjects who think they are first or second best, men enter more often.
- Table VI column (1) reproduces the baseline performance-controlled female effect in the restricted sample: −0.379.
- Column (2) adds guessed tournament rank. The guessed-rank coefficient is −0.181 and significant, so more confidence predicts entry.
- Adding guessed rank reduces the female effect from −0.379 to −0.278.
- The reduction means that about 27% of the gender gap is attributable to men being more overconfident, leaving about 73% unexplained by beliefs alone.

Interpretation: Confidence is important but incomplete. The fact that men enter more even conditional on guessed rank points toward additional factors beyond measured beliefs.

Economic message: Competitive institutions reward both ability and confidence. If confidence differs from ability, the tournament-entry pool can be distorted.

GENDER DIFFERENCES IN COMPETITIVENESS

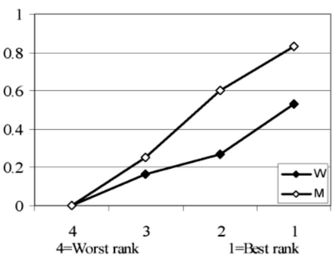

FIGURE III

(a) Figure III: Tournament entry conditional on guessed rank

To better understand the gender gap in tournament entry we need to consider the other explanations we proposed in Section II. We start by determining the effect general factors such as overconfidence and risk and feedback aversion have on the tourna-

TABLE VI
PROBIT OF TOURNAMENT-ENTRY DECISION (TASK 3)

	Coefficient (p-value)	
	(1)	(2)
Female	−.379	−.278
	(.01)	(.01)
Tournament	.015	−.002
	(.39)	(.90)
Tournament–piece rate	.008	−.001
	(.72)	(.94)
Guessed tournament rank		−.181
		(.01)

Dependent variable: Task-3 choice of compensation scheme (1-tournament and 0-piece rate). The table presents marginal effects evaluated at a man with a guess of one and who has thirteen correct answers in the tournament and twelve in the piece rate. Guesses of four are eliminated, resulting in a sample of thirty-eight women and thirty-nine men.

(b) Table VI: Probit of Task-3 tournament entry with guessed rank

5.6 Figure IV: Task-4 Tournament Submission; Table VII: Probit of Decision to Submit the Piece Rate to a Tournament

Read:

- Task 4 asks whether subjects want to submit their Task-1 piece-rate performance to a tournament.
- Because the performance already happened, the Task-4 decision does not require performing under pressure or competition.
- Figure IV compares submission rates by performance and guessed piece-rate rank. Beliefs strongly predict submission.
- Conditional on guessed rank, the Task-4 gender gap is much smaller than the Task-3 entry gap.
- Table VII column (1) shows a female marginal effect of -0.327 when controlling for piece-rate performance only.
- Column (2) adds guessed piece-rate rank; the female effect falls to -0.13 and is no longer significant ($p = .21$).
- The guessed piece-rate-rank coefficient is -0.32 and significant, meaning more confident subjects submit more often.

Interpretation: When the decision does not involve a future competitive performance, confidence explains most of the gender gap. This is why Task 4 is crucial for separating confidence from taste for competitive performance.

Economic message: Some of the entry gap is about beliefs and general tournament features. The residual Task-3 gap arises because active competitive performance is different from submitting a past score.

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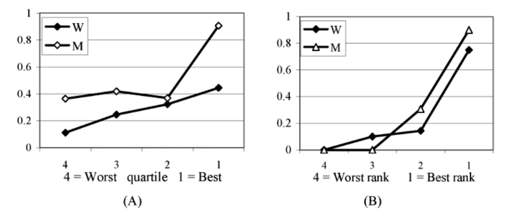

FIGURE IV
Proportion of Participants Who Select Tournament for Task 4 Conditional on Task-1 Performance Quartile (Panel A) and Guessed Piece-rate Rank (panel B).

(a) Figure IV: Task-4 submission by performance and belief

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TABLE VII
PROBIT OF DECISION TO SUBMIT THE PIECE RATE TO A TOURNAMENT (TASK 4)

	Coefficient (p-value)	
	(1)	(2)
Female	−.327	−.13
	(.01)	(.21)
Piece rate	.05	.00
	(.02)	(.80)
Guessed piece rate rank		−.32
		(.00)

Dependent variable: Task-4 choice of compensation scheme (1-tournament, 0-piece rate). The table presents marginal effects evaluated at a man with a guess of one and eleven correct answers in Task 1. Excluding guesses of 4, the sample is thirty-nine women and thirty-eight men.

(column (1)), this effect is to a large extent accounted for by

(b) Table VII: Probit of Task-4 submission decision

5.7 Table VIII: Probit of Tournament-Entry Decision with Beliefs and Task-4 Controls

Read:

- Table VIII returns to Task-3 tournament entry.
- Column (1) reports the baseline female effect of −0.379 with performance controls.
- Column (2) adds guessed tournament rank and reduces the female effect to −0.278.
- Column (3) adds Task-4 submission and reduces the female effect further to −0.162.
- The Task-4 submission coefficient is 0.258 and significant, so people willing to submit their past score are more likely to enter the active tournament.
- The guessed-rank coefficient remains significant in column (3), but it becomes smaller in magnitude, consistent with Task 4 partly capturing confidence.
- The remaining female effect is still significant ($p = .05$), suggesting that even after controlling for beliefs and general tournament-related factors, women are less likely to choose the active competitive performance.

Interpretation: Table VIII is the paper's central decomposition table. It shows how much of the gender gap is explained by beliefs and Task-4 behavior, and how much remains as a residual competitive-performance component.

Economic message: The result matters for real-world contests because applications, promotions, and bonuses often require active entry into competitive settings. Differences in willingness to enter can persist even after ability and confidence are accounted for.

TABLE VIII
PROBIT OF TOURNAMENT-ENTRY DECISION (TASK 3)

	Coefficient (p-value)		
	(1)	(2)	(3)
Female	−.379	−.278	−.162
	(.01)	(.01)	(.05)
Tournament	.015	−.002	−.009
	(.39)	(.90)	(.42)
Tournament–piece rate	.008	−.001	.011
	(.72)	(.94)	(.44)
Guessed tournament rank		−.181	−.120
		(.01)	(.01)
Submitting the piece rate			.258
			(.012)

Dependent variable: Task-3 compensation scheme choice (1-tournament and 0-piece rate). The table presents marginal effects evaluated at a man with thirteen correct answers in the tournament and twelve in the piece rate, who submits to the tournament, and with a guess of one in the Task-2 tournament. Guesses of four are eliminated, resulting in a sample of thirty-eight women and thirty-nine men.

Table VIII: Probit of Task-3 tournament-entry decision with beliefs and Task-4 controls

6 Short summary

- The paper studies whether equally able men and women differ in willingness to enter a tournament.
- Men and women perform similarly under both piece rate and compulsory tournament incentives.
- Despite similar performance, 73% of men and only 35% of women choose the tournament in Task 3.
- Performance controls and performance quartiles do not explain the gap.
- High-performing women often under-enter relative to expected-payoff benchmarks.
- Low-performing men often over-enter relative to expected-payoff benchmarks.
- Men are more overconfident: 75% of men guess they ranked best, versus 43% of women.
- Belief controls explain about 27% of the gender gap in Task-3 tournament entry.
- Task 4 shows that when no future competitive performance is required, gender differences are largely explained by beliefs.
- Adding beliefs and Task-4 choice explains about 57% of the original gender effect.
- The remaining 43% is interpreted as a residual difference in willingness to perform in competition.
- The findings imply that competitive institutions may affect entry and allocation even before performance evaluation occurs.

7 Conclusion

7.1 Core conclusion

The paper shows that gender differences in competition entry can arise even when measured ability is similar. Men and women do not simply perform differently in competitive environments; they choose whether to enter those environments differently. This distinction is central: if people self-select into tournaments, the observed set of competitors may not reflect the underlying distribution of ability.

7.2 Mechanism conclusion

The evidence points to a multi-channel mechanism. The first channel is confidence. Men are substantially more likely to believe that they ranked best, and belief controls explain part of the tournament-entry gap. The second channel is a general willingness to choose a tournament payoff, captured by Task 4. When the task is to submit a past score rather than perform again, confidence explains most of the gender difference. The third channel is the residual willingness to perform in an active competitive setting. This residual remains statistically and economically meaningful in Table VIII.

7.3 Interpretation of the 57%/43% decomposition

The decomposition should be read as a disciplined accounting exercise rather than a structural causal decomposition. Beliefs and Task-4 submission explain about 57% of the baseline female effect, while the remaining 43% is associated with active competitive performance. The residual could include dislike of competitive pressure, anxiety about performing under comparison, lower enjoyment of tournaments, stereotype-related concerns, or other unmeasured factors not fully captured by Task 4.

7.4 Economic intuition

A tournament rewards the winner, but only among people who enter. A high-ability person who does not enter cannot win, while a low-ability person who enters can still crowd the contest or receive inefficient expected payoffs. The experiment shows both sides. Some women avoid profitable tournaments, and some men enter unprofitable tournaments. This implies that tournament-based institutions may misallocate opportunities even when their scoring rule is unbiased.

7.5 Implications for labor markets and organizations

Many labor-market outcomes depend on voluntary entry into competitive processes: applying for promotions, choosing performance pay, negotiating, entering elite tracks, running for leadership, or applying to competitive programs. If entry preferences differ by gender, then organizations can observe gender gaps in winners even without gender gaps in ability. The paper suggests that changing the entry environment - for example by providing clearer information about relative performance, reducing unnecessary winner-take-all features, or encouraging qualified candidates to apply - may affect both efficiency and diversity.

7.6 Policy implications

The results do not imply that competition should be eliminated. Tournaments can create incentives and select high performers. The implication is more subtle: organizations should ask whether the decision to enter the tournament is itself screening for the desired trait or screening for a separate taste for competition. If the goal is to select ability, overly competitive entry procedures may miss qualified individuals. If the goal is to select willingness to compete, the tournament is doing exactly that but should be interpreted accordingly.

7.7 Limitations

The experiment is clean but stylized. The task is short, individual, and arithmetic-based. Real careers involve repeated interactions, teamwork, strategic behavior, discrimination, networks, family constraints, experience, and institutional rules. The sample is small, and the findings should not be interpreted as universal for all women or all men. Task 4 is a clever control, but it may not fully capture every component of risk aversion, feedback aversion, or confidence.

7.8 Broader takeaway

The paper's contribution is to make self-selection into competition an economic mechanism. Gender gaps can emerge before employers, schools, or markets evaluate performance, because people differ in whether they choose to enter the competitive arena. The central lesson is that competitiveness is not merely a personality trait; it is a decision margin that can shape allocation, inequality, and institutional outcomes.